Calculus I Lesson 43

1) a) Evaluate $\int_{1}^{2} \frac{1}{x} \cdot dx =$ ___?___ Hint: Use the Area Under Curve Program

b) Evaluate $\int_{1}^{2.7} \frac{1}{x} \cdot dx =$ ___?___

c) Evaluate $\int_{1}^{2.7182} \frac{1}{x} \cdot dx =$ ___?___

d) Evaluate $\int_{1}^{2.718281828459} \frac{1}{x} \cdot dx =$ ___?___

e) Evaluate $\int_{1}^{2.718281828459045} \frac{1}{x} \cdot dx =$ ___?___

2) Solve $e^x = 15$.

Hint: $e^{\ln u} = u$; $\ln e^u = u$; $\ln x = \log_e x$; $\ln e = 1$

$x =$ ___?___

3) Solve $19 - 2e^{x} = 4$.

Hint: $e^{\ln u} = u$; $\ln e^{u} = u$; $\ln x = \log_{e} x$; $\ln e = 1$

$x =$ ___?___

4) Solve $10e^{-x} = 30$.

Hint: $e^{\ln u} = u$; $\ln e^{u} = u$; $\ln x = \log_e x$; $\ln e = 1$

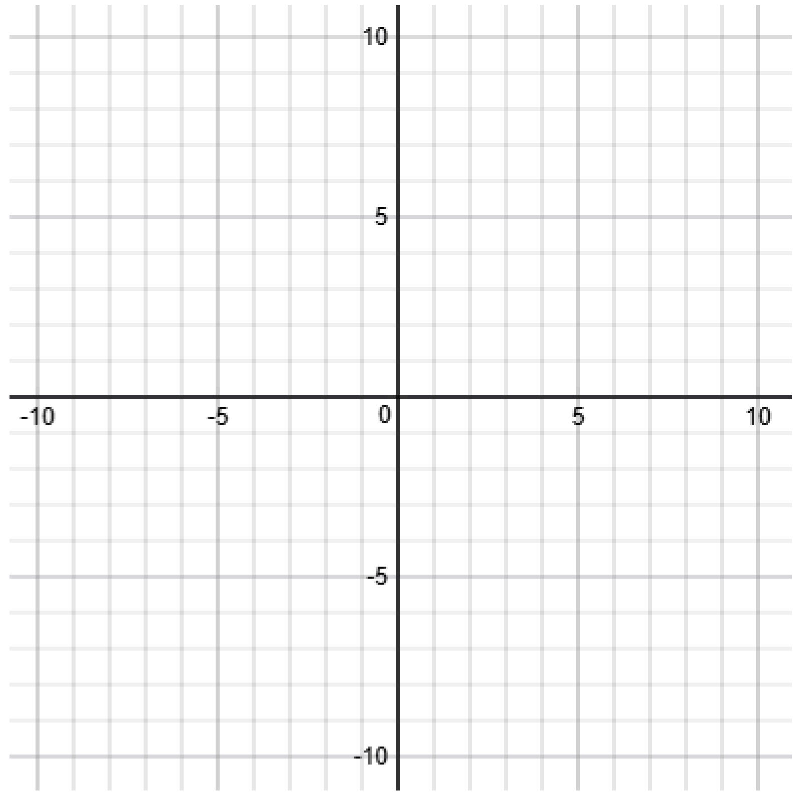
$x =$ ___?___

5) $f(x) = e^x + 2$

a) Domain of $f(x) =$ ___?___

b) Range of $f(x) =$ '___?___

c) Graph $f(x)$.

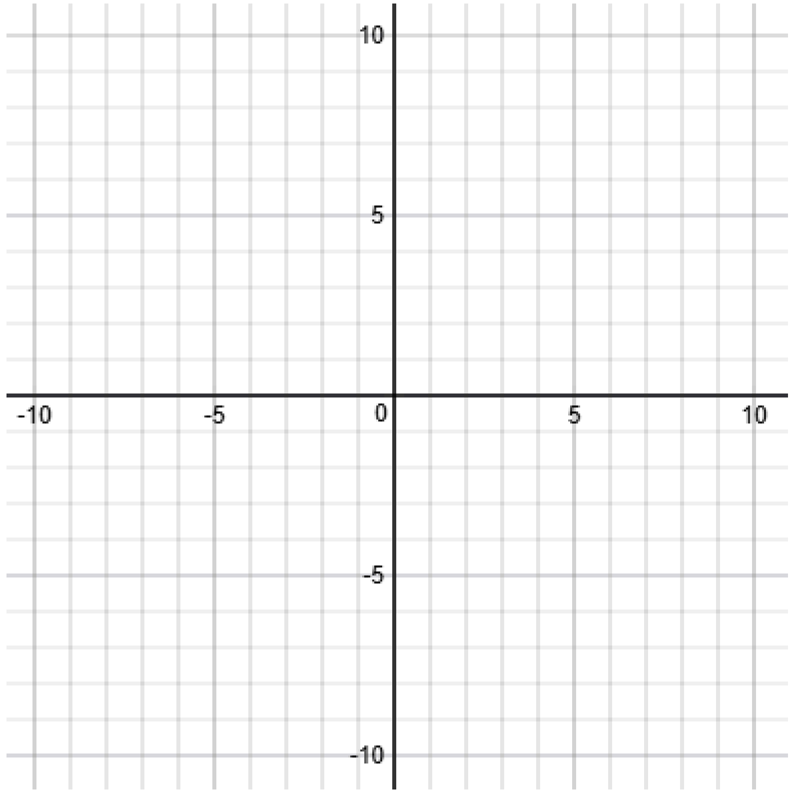


6) $f(x) = e^{-2x^2}$

a) Domain of $f(x) =$ ___?___

b) Range of $f(x) =$ ___?___

c) Graph $f(x)$.



7) a) $\ln e^{2} =$ ___?___ $\qquad e^{\ln 2} =$ ___?___

b) $\ln e^{3} =$ ___?___ $\qquad e^{\ln 3} =$ ___?___

c) $\ln e^{5} =$ ___?___ $\qquad e^{\ln 5} =$ ___?___

d) $\ln e^{x} =$ ___?___ $\qquad e^{\ln x} =$ ___?___

8) Note: $4 = e^{\ln 4}$; $7 = e^{\ln 7}$; $15 = e^{\ln 15}$

$b = e^{\ln b}$; $c = e^{\ln c}$; $d = e^{\ln d}$

So $a^x = e^{\ln a^x} = e^{x \cdot \ln a}$

Find $D_x\left(a^x\right) = D_x\left(e^{x \cdot \ln a}\right) = \underline{\quad ? \quad}$

Hint: Let $u = x \cdot \ln a$ and use the Chain Rule $D_x\left(e^u\right) = e^u \cdot \dfrac{du}{dx}$